

Basketball Jump Shooting Is Controlled Online by Vision

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Abstract. An experiment was conducted to examine whether basketball jump shooting relies on online visual (i.e., dorsal stream-mediated) control rather than motor preprogramming. Seventeen expert basketball players (eight males and nine females) performed jump shots under normal vision and in three conditions in which movement initiation was delayed by zero, one, or two seconds relative to viewing the basket. Shots were evaluated in terms of both outcome and execution measures. Even though most shots still landed near the basket in the absence of vision, end-point accuracy was significantly better under normal visual conditions than under the delay conditions, where players tended to undershoot the basket. In addition, an overall decrease of inter-joint coordination strength and stability was found as a function of visual condition. Although these results do not exclude a role of motor preprogramming, they demonstrate that visual sensory information plays an important role in the continuous guidance of the basketball jump shot.

Keywords: basketball shooting, visual control, expert performance, far aiming

How far-aiming movements like kicking or throwing a ball are organized and guided by visual information has become an important research theme of late. With regard to the question when visual information is used in motor control, an old dichotomy has recently surfaced in studies of basketball shooting. While some researchers have emphasized that aiming movements are preprogrammed before execution, other researchers have argued for their online control, dismissing the need for detailed motor programming. For instance, Vickers (1996) found that, when preparing a basketball free throw, players fixated their gaze on the target for a long time before initiating the shot, and that gaze fixation was terminated once they started the shot phase of the movement (i.e., final elbow extension). She proposed that players needed visual information before the shot to preprogram movement parameters, allowing the shot to be performed without sensory feedback. In contrast, Oudejans, van de Langenberg, and Hutter (2002) found that basketball players could accurately perform a jump shot when their vision was blocked up to the last 350 ms before ball release, whereas performance was hampered when vision was blocked during this period (which includes the final elbow extension). Oudejans et al. concluded that having only online visual information available during the shot is sufficient for accurate performance, rendering elaborate motor preprogramming superfluous.

These contrasting results and interpretations concerning the classic preprogramming-online control dichotomy may be understood in view of recent neurophysiological in-

sights into the visual control of action. In this research, temporal aspects of motor control have been linked to the distinction between the dorsal and ventral streams of visual information processing (cf. Milner & Goodale, 1995; Rossetti & Pisella, 2002). According to this dichotomy, the dorsal stream uses visual information coded in motor coordinates for the continuous guidance of movement, whereas the ventral stream uses visual information for object identification (Milner & Goodale, 1995). Initial neurological evidence for the dorsal-ventral distinction came from studies on DF, a patient with visual form agnosia (i.e., damage of the ventral pathway), who could accurately reach and grasp objects while being unable to describe or identify such objects. In contrast, AT, a patient with optic ataxia (i.e., damage of the dorsal pathway) could describe objects, but showed poor performance when asked to point to objects, suggesting that she was unable to properly guide her movements online. In contrast to healthy individuals, AT's pointing errors reduced when a delay was introduced between viewing the object and making the pointing movement (Milner et al., 2001; Milner, Paulignan, Dijkerman, Michel, & Jeannerod, 1999). These results have led to the insight that dorsal stream information is used for the online control of action (even if the processing of this information is compromised by brain damage), while ventral stream information may come into play when movement execution is delayed relative to the timing of visual information pick-up (Pisella, et al., 2000; Rossetti & Pisella, 2002).

Delaying the movement relative to the pick-up of visual

information thus constitutes a promising behavioral manipulation to study the visual guidance of basketball shooting (Rossetti & Pisella, 2002), which may cast light on the contrasting results that have been published so far on this topic. Since the dorsal system is devoted primarily to the online guidance of movements, this stream of visual information processing can be disrupted by introducing a visual delay. In contrast, the ventral system is dedicated to longer-term cognitive processes, and will thus remain functional after relatively long visual delays. It has been established that the briefest delay after which changes in motor control are observed is 500 ms and that a 2-s delay is long enough to produce observable changes in motor output (Rossetti & Pisella, 2002).

To examine whether basketball jump shooting relies on the online (i.e., dorsal stream-mediated) use of visual information, we had expert basketball players perform jump shots under normal visual conditions and conditions in which the shooting movement was delayed relative to the availability of vision by zero, one and two seconds. Motor performance was evaluated in terms of end-point accuracy and the kinematics of the shooting arm (see Method). Players with low and high shooting styles were included in the experiment in view of Oudejans et al.'s (2002) suggestion that the visual control of basketball jump shooting depends on the adopted shooting style. This suggestion was based on the fact that the low shooting style involves simultaneous shoulder anteflexion and elbow extension, resulting in an upward and forward pushing movement that brings ball and hands into the shooter's field of view thereby occluding the basket just before ball release, whereas the high shooting style is characterized by a shoulder anteflexion that brings the ball above the head, followed by an elbow extension that propels the ball toward the basket, allowing

vision of the basket throughout the movement. Oudejans et al. (2002) therefore conjectured that low-style shooters would preprogram their movement before execution, whereas high-style shooters would rely solely on online control. According to this hypothesis, high-style shooters would be expected to show marked decrements in performance under the visual delay conditions, whereas low style shooters would be expected to be less susceptible to (externally) imposed visual delays, provided they are brief. After all, under the low style the shooting movement itself already introduces a visual delay by virtue of its kinematic characteristics, whereas under the high style the target is always visible. In short, if basketball players preprogram their shooting actions, as might be the case under the low style, then a brief visual delay should not interfere with performance. Conversely, if online control is used, as might be the case under the high style, then brief visual delays should lead to marked decrements in performance. The experiment reported here was designed to examine these expectations.

Method

Participants

Seventeen experienced basketball players (mean age = 23.9 years, $SD = 5.8$ years) participated in the study. All played either at the guard or forward position in the two highest leagues in The Netherlands and had 10.4 years of experience on average ($SD = 6.0$). Nine players had a high shooting style (eight male and one female) and eight a low shooting style (all female), which was confirmed after the experiment through video-footage analysis using previously established criteria (de Oliveira, Oudejans, & Beek, 2006; Oudejans et al., 2002). There were no significant ($p > .05$) group differences regarding age or experience ($t_{15} = 1.28$ and $t_{15} = 0.09$, respectively). The experiment was approved by the local ethics committee of the Faculty of Human Movement Sciences.

Experimental Setup

A standard basketball backboard and rim were set up in a large laboratory and a shooting area was defined about 4.6 m away from it (see Figure 1). To the right of the shooting area two Optotrak sensor units (Optotrak 3020, Northern Digital Inc., Waterloo, Canada; sampling rate 100 Hz) were placed to capture the shooters' movements. In addition, two digital video cameras were placed on both ends of the setup to record the entire ball trajectory at 25 Hz. Light emitting diodes that indicated visual condition, start cue (delivered via an earphone to the shooter), and ball release (captured by a micro-switch) were placed in view of the cameras.

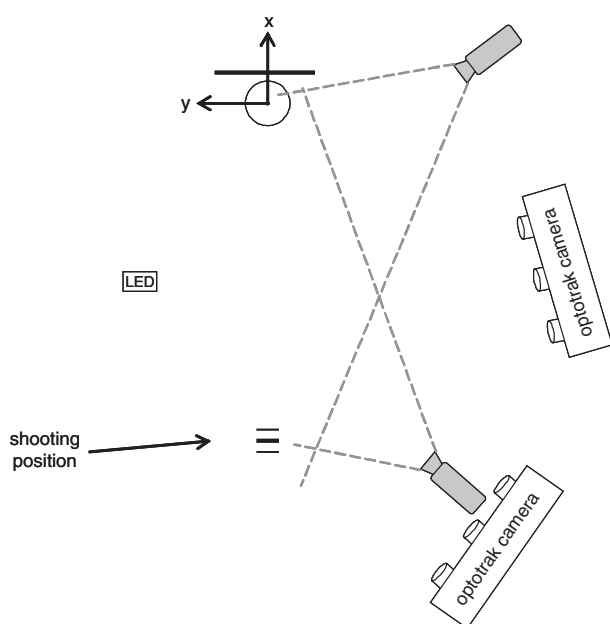


Figure 1. Top view of the experimental setup (see text).

Participants wore Plato LC glasses (Translucent Technologies, Toronto, Canada) that could switch from transparent to opaque, thus enabling and preventing vision. Nine Optotrak active infrared markers were recorded; six markers were attached to the participants' shooting arm (little finger, ring finger, metacarpal area, wrist, elbow, and shoulder), and three others were taped to the right leg of the glasses. A micro-switch was taped to the shooters' middle finger to detect the moment of ball release, which triggered a masking sound (white noise superimposed on a 100 Hz frequency) on an earphone with 0.5 s delay so as to prevent auditory feedback from the ball's landing position. The earphone also cued the start of each trial. Official regulation-sized, FIBA basketballs (Spalding) were used.

Design

Participants performed 12 blocks of 12 trials for a total of 144 jump shots. The shots were taken under four visual conditions, randomized over each 12-trial block: A full-vision condition in which the glasses remained open (V_f); a 0-s delay condition in which the glasses closed at the cue to shoot (V_0); a 1-s delay condition in which the cue to shoot was given 1 s after the glasses had closed (V_1); and a 2-s delay condition in which the cue to shoot was given 2 s after the glasses had closed (V_2). To ensure that players would not shoot each time from the same position, they were instructed to shoot randomly from three distances (4.3, 4.6, and 4.9 m), which were marked on the floor. In addition, the backboard was positioned alternately either parallel or orthogonal to the player's sagittal plane on each trial block.

Procedure

The participant gave his or her written informed consent after a brief explanation of the purpose of the experiment and the experimental procedure. The experimenters placed the Optotrak markers, glasses, micro-switch, and earphone. Participants took several shots to warm-up and get used to the experimental set-up. They were instructed to look at the basket for 3 seconds on each trial, to initiate their movements immediately after the signal and to shoot at their own pace. During the initial 3 s players were free to stay still or bounce the ball. During each trial visual feedback from the ball trajectory was prevented by closing the glasses 0.05 s after ball release, while auditory feedback was eliminated by delivering the masking sound over the earphone 0.5 s after ball release. After each trial the result of the shot was registered as hit or miss, the glasses opened, and the ball was returned to the player.

Data Processing

The coordinate system was defined with the x-axis increasing from the starting position to the basket, the y-axis increasing leftwards, and the z-axis increasing upward, such that the center of the rim was $x = 0$, $y = 0$, $z = 3050$ mm. We calculated the percentage of hits for each player and submitted this variable to a repeated measures ANOVA with within-subjects factor vision (4 levels) and between-subjects factor group (2 levels). To augment the discriminatory power of the performance measure, we also examined the landing positions of the ball relative to the center of the rim along the x- and y-axis. This was accomplished by digitization (using WinAnalyse, Mikromak) of the ball's flights in 3D, followed by a fitting procedure that allowed estimation of the ball's position when its center (would have) crossed the horizontal plane of the rim. This procedure was necessary as the ball often hit the rim or backboard before the ball's center crossed the plane of the rim. Because the 3D digitization of the ball trajectories was laborious, this analysis was restricted to a random sub-sample of 10 participants. There were no significant ($p > .05$) differences between the five high-style and the five low-style shooters in this subsample with respect to age or experience ($t_{5,8} = 1.28$ and $t_8 = 0.25$, respectively). From the estimated landing positions of the ball on the (horizontal) plane of the rim, we calculated the average absolute error along the x- and y-axis as well as the corresponding standard deviations (both measures in mm). To examine the effects of the visual conditions, we submitted both absolute errors and standard deviations to a repeated measures MANOVA on x- and y-axis, with within-subjects factor vision (4 levels) and between-subjects factor group (2 levels).

The raw kinematic data of 14 participants were filtered using a fourth-order, low-pass Butterworth filter with a cut-off frequency of 14 Hz and subsequently normalized to unit variance. Data from three participants were excluded from this analysis because they contained lacunae due to occlusion of the markers. No significant ($p > .05$) differences were found between the remaining six high-style and eight low-style shooters with respect to age or experience ($t_{12} = 1.24$ and $t_{12} = 0.36$, respectively). Next, we calculated the covariance function between the following joint pairs: Shoulder-elbow, elbow-wrist, and wrist-finger in the x- and z-axis (as the shooting movement evolved primarily in the xz-plane), determined the time lags at which the covariance was maximal, and calculated the intra-individual means and standard deviations of the covariance coefficients at lag zero. The effect of the visual conditions on the strength and variability of inter-joint coordination, or coupling, was examined by submitting the means and standard deviations of the covariance coefficients in the x- and z-axis to repeated measures MANOVAs on the joint-pairs shoulder-elbow, elbow-wrist, and wrist-finger, again with within-subjects factor vision (4 levels) and between-subjects factor group (2 levels). The exact statistics are reported for the multivar-



Figure 2. Distribution of the balls' landing positions relative to the rim of the basket under full vision (V_f), 0-s delay (V_0), 1-s delay (V_1), and 2-s delay (V_2). Each dot represents the position of the center of the ball when it crossed the horizontal plane of the rim. Circles represent the rim, which had a diameter of 450 mm. Data of two high-style shooters (12H and 14H) and two low-style shooters (13L and 19L) are plotted. The ball flights evolved along the x-axis. Note the marked undershooting of the rim under the delay conditions.

iate tests using Wilk's Lambda. Where appropriate, the degrees of freedom were adjusted for violations of sphericity using the Huynh-Feldt correction. Significant main effects were examined further using pairwise comparisons with the Bonferroni correction procedure.

Results

Shooting Accuracy

There was a significant effect of vision, $F(3, 45) = 10.03$, $p < .001$, $\eta^2 = .40$, revealing that the percentage of hits was significantly larger in the full-vision condition (V_f : $M = 32.7\%$, $se = 4.9$; all p values $< .05$) than in the visual delay conditions, which were not significantly different from each other (V_0 : $M = 17.0\%$, $se = 4.1$; V_1 : $M = 15.9\%$, $se = 3.9$; V_2 : $M = 18.3\%$, $se = 4.7$; all p values $> .05$). The absence of a significant Vision $\times$ Group interaction or a group effect indicated that shooting accuracy was independent of shooting style. These results were preserved when the ANOVA was restricted to the subsample of 10 participants selected for the analysis of the landing position, $F(3, 24) = 11.01$, $p < .001$, $\eta^2 = .58$, as well as the subsample of 14

participants included in the kinematic analysis, $F(3, 36) = 10.12$, $p < .001$, $\eta^2 = .46$.

Furthermore, there was a significant effect of vision, $F(6, 46) = 4.17$, $p < .01$, $\eta^2 = .35$, on the absolute error of the ball's landing position along both the x-axis, $F(1.4, 11.1) = 6.67$, $p < .05$, $\eta^2 = .46$, and the y-axis, $F(2.2, 17.8) = 3.57$, $p < .05$, $\eta^2 = .31$. Along the x-axis, the absolute error was smaller under the full-vision (V_f : $M = 175$, $se = 18$) than under the delay conditions (V_0 : $M = 288$, $se = 68$; V_1 : $M = 306$, $se = 64$; V_2 : $M = 304$, $se = 60$). The same effect occurred along the y-axis (V_f : $M = 97$, $se = 10$ versus V_0 : $M = 116$, $se = 15$; V_1 : $M = 128$, $se = 21$; V_2 : $M = 122$, $se = 17$). The distribution of the ball's landing positions revealed that shooting errors consisted predominantly of undershooting the center of the rim (see Figure 2).

Finally, there was a significant effect of vision on the standard deviation of the error, $F(6, 46) = 5.21$, $p < .001$, $\eta^2 = .40$, that was also present for both the x-axis, $F(3, 24) = 3.34$, $p < .05$, $\eta^2 = .29$, and the y-axis, $F(3, 24) = 9.13$, $p < .001$, $\eta^2 = .53$. Variability increased from the full-vision condition to the 2-s delay condition and was larger along the x-axis (V_f : $M = 163$, $se = 14$ versus V_0 : $M = 161$, $se = 13$; V_1 : $M = 188$, $se = 15$; V_2 : $M = 205$, $se = 14$, with V_f significantly smaller than V_2 ; $p < .05$) than along the

y-axis (V_f : $M = 98$, $se = 6$ versus V_0 : $M = 117$, $se = 7$; V_1 : $M = 123$, $se = 8$; V_2 : $M = 130$, $se = 7$, with V_f and V_1 significantly smaller than V_2 ; $p < .05$).

In sum, the visual manipulation had a marked effect on shooting accuracy and its variability, in the absence of significant Vision $\times$ Group interactions or group effects ($p > .05$). The percentage of hits was substantially smaller under the visual delay conditions than under the full-vision condition. In fact, the distance between the balls' landing positions and the center of the rim was larger under the visual delay conditions than under full vision. This effect was accompanied by larger variability under the three delay conditions along both the x- and y-axis. The error distribution revealed that players tended to undershoot the basket under the visual delay conditions.

Interjoint Coordination

The time lag for which the covariance coefficient was largest was close to zero for all joint pairs (i.e., the shoulder-elbow, elbow-wrist, and wrist-finger joint pairs) along both the x- and z-axis. The coefficients were, in general, positive, implying that the joints moved in the same direction along these axes. Because the time lags were only marginally different from zero (if at all), the covariance coefficient at zero time-lag was used in the statistical analyses. On average, coupling strengths along the x-axis were smaller in the shoulder-elbow ($M = .66$, $SD = .30$) and elbow-wrist joint pair ($M = .44$, $SD = .27$) than in the wrist-finger joint pair ($M = .82$, $SD = .15$), whereas average coupling strengths along the z-axis were larger than .93 in all joint pairs.

Along the x-axis, there was a significant overall effect of vision on the covariance coefficients, $F(9, 82.9) = 2.58$, $p < .05$, $\eta^2 = .18$, that occurred predominantly in the elbow-wrist coupling, $F(3, 36) = 4.06$, $p < .05$, $\eta^2 = .25$. As the severity of the imposed visual constraints increased, the degree of elbow-wrist coupling decreased (V_f : $M = .22$, $se = .14$ versus V_0 : $M = .18$, $se = .14$; V_1 : $M = .16$, $se = .15$; V_2 : $M = .14$, $se = .14$). As regards the standard variations of these covariance coefficients, we found a significant effect of vision on shoulder-elbow coupling, $F(3, 36) = 3.30$, $p < .05$, $\eta^2 = .22$, where variability was largest under the 2-s delay condition (V_f : $M = .22$, $se = .04$; V_0 : $M = .20$, $se = .04$; V_1 : $M = .21$, $se = .05$; V_2 : $M = .26$, $se = .05$). We found the same tendency for the elbow-wrist joint pair, $F(3, 36) = 3.34$, $p = .09$, $\eta^2 = .16$, again with the largest variability under the 2-s delay (V_f : $M = .25$, $se = .04$; V_0 : $M = .22$, $se = .02$; V_1 : $M = .23$, $se = .03$; V_2 : $M = .28$, $se = .03$). These results indicate that movement execution was least consistent in the 2-s delay condition. No significant Vision $\times$ Group interactions or group effects were found along the x-axis for average or standard deviation of the covariance coefficients.

Along the z-axis, there was a significant overall effect of vision, $F(9, 90.2) = 2.64$, $p < .01$, $\eta^2 = .17$, that occurred predominantly in the shoulder-elbow coupling, $F(2.0, 26.2) =$

3.78 , $p < .05$, $\eta^2 = .23$, even though the degree of this coupling decreased only minimally with increasing severity of visual conditions (V_f : $M = .920$, $se = .02$; V_0 : $M = .922$, $se = .02$; V_1 : $M = .919$, $se = .02$; V_2 : $M = .911$, $se = .02$). No significant effects of group were found, but there was a tendency toward a significant Vision $\times$ Group interaction for the elbow-wrist pair, $F(2.5, 32.2) = 2.54$, $p = .08$, $\eta^2 = .16$, in that coupling strength under the 0-s delay condition decreased in the high-style group and increased in the low-style group. As regards coordination variability, there was a significant overall effect of vision, $F(9, 82.9) = 3.74$, $p < .01$, $\eta^2 = .24$, that was most pronounced for the shoulder-elbow pair, $F(1.7, 20.1) = 9.36$, $p < .01$, $\eta^2 = .44$. Again, the variability of this joint linkage was largest under the 2-s delay condition (V_f : $M = .028$, $se = .01$; V_0 : $M = .026$, $se = .01$; V_1 : $M = .024$, $se = .01$; V_2 : $M = .045$, $se = .01$). In addition, there was a marginal group effect for the elbow-wrist pair, $F(1, 12) = 3.97$, $p = .07$, $\eta^2 = .25$, because the overall variability of inter-joint coordination tended to be larger in the high-style group ($M = .02$, $se = .004$) than in the low-style group ($M = .01$, $se = .003$). No significant Vision $\times$ Group interaction was present.

In sum, these results showed that the severity of the imposed visual constraints was associated with a decrease in coupling strength between the arm joints, especially for the elbow-wrist pair along the x-axis and the shoulder-elbow pair along the z-axis. Increasing the severity of the imposed visual constraints further resulted in increased movement variability, particularly so for the shoulder-elbow pair.

Discussion

In the present study, we examined the nature of the visual control of basketball jump shooting in terms of the dichotomy between preprogramming and online control. Based on previous suggestions (Oudejans et al., 2002; Vickers, 1996), we hypothesized that players with a low style would preprogram their movements, whereas players with a high style would control their movements online. Expert players of both styles performed jump shots with full vision and under visual delays of zero, one or two seconds. We expected high-style shooters to show large decrements in performance under the visual delay conditions, whereas low-style shooters were expected to be less susceptible to (externally) imposed visual delays of brief duration (i.e., under V_0 but not under V_1 and V_2).

We evaluated shooting accuracy in terms of the percentage of hits as well as the average and variability of the ball's landing position relative to the center of the rim. In all three measures performance was significantly better in the full vision condition than in the three visual delay conditions (in the absence of significant differences among them). Contrary to our expectation, there was no significant Delay $\times$ Group interaction, indicating that the low-style shooters did not employ a different kind of control than the high-style shooters. Although low-style shooters indeed occlud-

ed their vision of the basket during the final elbow extension, the duration of this occlusion was only 123 ms in the two participants in Oudejans et al. (2002), which is well within the visual-motor delays reported in the literature. Thus, in principle, low-style shooters could control the shooting movement in the same way as high-style shooters (i.e., online control), implying that, in hindsight, the association of shooting style with type of motor control underlying our hypothesis might have been invalid.

The conclusion that low-style and high-style shooters in fact used the same kind of control is consistent with the results of de Oliveira et al. (2006), who studied the preferred timing of optical information pick-up in shooters of both styles using an intermittent vision paradigm. In the experiment in question viewing windows of 350 ms duration were applied to eliminate the possibility of extensive preprogramming. In spite of this, both high-style and low-style shooters preserved their full-vision shooting accuracy. Furthermore, shooters with either style picked up optical information as late as possible as permitted by the adopted shooting style, that is, until the moment of ball release itself in the high-style shooters and until ball and hand started occluding the target in the low-style shooters (i.e., a fraction of a second before actual ball release). Thus, in combination with the present results, it seems that the accuracy of expert players with either style derives from an optimal use of online information. This general conclusion, however, does not imply that motor programming played no role whatsoever in the organization of the jump shot. Even in the conditions in which movement initiation was delayed for 1 or 2 s, all shots landed in the vicinity of the basket (< 1 m), indicating that in those conditions shooting was not random but that the shooting movement was programmed to some degree when the basket was visible. However, from the fact that shooting performance deteriorated with the delay in movement initiation, we might conclude that remembering the exact location of the basket in a somewhat far, extrinsic space is a rather demanding task. It may well be that this limitation in the working memory capacity for remembered target location forces expert basketball players to rely on visual information as long as it is available, allowing them to tune the action online to the basket. This interpretation is consistent with recent suggestions in the literature that actions are organized “effortlessly” so as to achieve perceptible goal states (Mechsner, 2004).

The distribution and variability of endpoint accuracy revealed undershooting errors in the anterior-posterior direction (x-axis) that were markedly larger under the three delay conditions than under full vision. This is in line with past research on near aiming showing that errors are typically observed in the movement direction, especially when vision is restricted (Khan, Lawrence, Franks, & Buckolz, 2004), as well as studies on reaching that have found undershooting effects under visual delay conditions (Westwood, Heath, & Roy, 2003). The undershooting observed in the present study may have been caused by players misperceiving the occluded basket as being closer to their shooting position. In studying eye-movements to remem-

bered visual targets, Gnadt, Bracewell, and Andersen (1991) found upward shifts in the endpoint of saccades to occluded targets as opposed to visible targets. If basketball players indeed misperceived the occluded basket as being higher in their visual field, then they may have perceived it as being closer (see Oudejans, Koedijker, Bleijendaal, & Bakker, 2005). Given that people use gaze to guide goal-directed movements (Soechting, Engel, & Flanders, 2001; Vickers, 1996), shooting at a basket being misperceived as closer would cause undershooting it.

We also examined the consequences of imposing delays on movement coordination by investigating the strength and variability of the coupling between joints in the xz-plane. We found that all joint pairs were coupled strongly along the z-axis in all delay conditions and in both groups, indicating that all joint pairs were implicated in the movement. This result is manifest in view of the fact that the entire movement pattern was one of transporting the ball upward to its release point. Along the x-axis we found weaker couplings for the shoulder-elbow and elbow-wrist pairs. Furthermore, we found large coupling strengths at the wrist-finger joint pair along both the x- and z-axis in both groups under all conditions. This finding is readily understandable from the kinematics of the unfolding action in that the movements of the hand and fingers are severely constrained by carrying the ball, requiring a continued dorsal flexion of the hand. Hence, the strong couplings and small variability for the wrist-finger joint pair reflect a largely fixed relative position of wrist and finger that was unaffected by the visual conditions.

Importantly, there was a significant effect of the vision manipulation on the coupling strength as well as on its variability in the shoulder-elbow and elbow-wrist joint pairs along both the x- and z-axis. Overall, inter-joint coordination strength and stability decreased as a function of visual condition. This effect was especially pronounced in the 2-s delay condition, i.e., the most severe of all visual delays. This effect suggests that visual information played a role in the continuous guidance of the basketball jump shot, favoring interpretations of online control rather than motor preprogramming. Notwithstanding earlier suggestions that sensory information cannot be used in the online control of rapid (hand) movements (cf. Jeannerod, 1988), the view that fast aiming movements can be controlled online on the basis of visual information is rapidly gaining ground, particularly in studies of forward models of motor control. Forward models propose a feedback system that uses sensory information about the visual target position as well as motor-related changes and motor output to estimate hand location. An error signal is generated by comparing the visual target position and the internal estimate of hand location that will be used to perform the necessary adjustments to the ongoing movement (see Desmurget et al., 1999). In the delay conditions of the present study visual target location information was not available, implying that the error signal for movement guidance may have been generated on the basis of an estimate of hand and arm position as well as gaze position (including its up-shift bias as dis-

cussed in the preceding). The consequence for the movement kinematics was a decrease in the reproducibility of actions along with weaker couplings for the shoulder-elbow and elbow-wrist joint pairs.

The brain area thought to be involved in the error signal generation (i.e., the instantaneous difference between hand and target position) is the posterior parietal cortex, which is part of the dorsal stream of visual information processing. Previous studies have shown that lesion or disruption of the posterior parietal cortex (by transcranial magnetic stimulation) prevent movement adjustments toward targets if their location was changed unbeknownst to participants after movement onset. Evidence from healthy participants (Desmurget et al., 1999) as well as patients with lesions in this brain area (Grea et al., 2002) underscores the role of this structure in the control of online adjustments during aiming movements. Desmurget et al. (1999) have shown that disruption to posterior parietal cortex activity prevents healthy participants from correcting their movements toward targets if these targets are moved but not if they remain stationary. Other authors have also shown that participants perform adjustments to their hand trajectory when targets are displaced unbeknownst to them (Prablanc & Martin, 1992). The present results demonstrate that the online control of movement based on visual information about the target also prevails in dynamic perceptual-motor tasks like basketball jump shooting.

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